

CUSTOMER ANALYTICS AND LOYALTY ANALYSIS USING PYTHON, SQL, AND POWER BI

Executive Summary

This project analyzes customer purchasing behavior using a retail shopping dataset containing 3,900 customer records. The objective was to identify valuable customers, understand customer loyalty, analyze promotional dependency, evaluate customer satisfaction, and generate business recommendations that can improve customer retention and revenue growth.

The project was completed using Python for data cleaning and feature engineering, MySQL for business analysis, and Power BI for dashboard development. Multiple customer segmentation techniques were implemented, including Customer Value Score, Loyalty Score, Satisfaction Analysis, and Promotional Dependency Analysis.

The findings revealed that Clothing is the highest revenue-generating category, Medium Value customers represent the largest customer segment, and approximately 43% of customers are highly dependent on promotions. The analysis also identified an ideal customer segment consisting of high-value, highly loyal customers with low promotional dependency.

1. Introduction

Customer analytics plays a critical role in helping businesses understand purchasing behavior, customer preferences, and loyalty patterns. By analyzing customer data, organizations can improve marketing effectiveness, increase customer retention, and maximize revenue.

This project focuses on customer segmentation and loyalty analysis using historical retail transaction data. The analysis aims to uncover meaningful business insights that can support strategic decision-making.

2. Business Problem

Retail companies often struggle to answer important business questions such as:

- Which customers generate the most revenue?
- Which customers are most loyal?
- How dependent are customers on discounts and promotions?
- Which product categories drive revenue growth?

- How can customer retention be improved?

This project addresses these questions through data analysis and visualization.

3. Dataset Overview

The dataset contains customer demographic information, purchase history, and shopping behavior attributes.

Dataset Statistics

- Total Records: 3,900
- Total Features: 18
- Missing Values: 37 (Review Rating)
- Duplicate Records: 0

Important Variables

- Customer ID
 - Age
 - Gender
 - Category
 - Purchase Amount (USD)
 - Location
 - Season
 - Review Rating
 - Subscription Status
 - Discount Applied
 - Promo Code Used
 - Previous Purchases
 - Payment Method
 - Frequency of Purchases
-

4. Data Cleaning

Several preprocessing steps were performed to ensure data quality.

Missing Values

The Review Rating column contained 37 missing values.

These missing values were replaced using the median review rating value.

Duplicate Records

No duplicate records were identified.

Data Validation

All data types were verified and validated before proceeding with analysis.

5. Exploratory Data Analysis

Customer Demographics

- Average Customer Age: 44 years
- Male Customers: 2,652
- Female Customers: 1,248

Male customers represent approximately 68% of the customer base.

Revenue Analysis

- Total Revenue: \$233,081
- Average Purchase Amount: \$59.76

Revenue by Gender

- Male Revenue: \$157,890
- Female Revenue: \$75,191

Male customers contribute significantly more revenue.

Revenue by Category

Category	Revenue
Clothing	\$104,264
Accessories	\$74,200
Footwear	\$36,093
Outerwear	\$18,524

Clothing is the highest-performing category.

Discount Analysis

Average spending remains nearly identical regardless of discount availability, indicating that discounts do not significantly increase purchase value.

Subscription Analysis

Subscribers and non-subscribers exhibit similar spending behavior, suggesting that the current subscription program has limited impact on purchase value.

6. Feature Engineering

To improve customer segmentation, several new business-oriented features were created.

Customer Value Score

Customer Value Score was calculated using:

Customer Value Score = Purchase Amount $\times$ Previous Purchases

Customers were classified into:

- High Value
- Medium Value
- Low Value

Promo Dependency Score

Promo Dependency Score was developed using:

- Discount Applied
- Promo Code Used

Customers were segmented into:

- High Dependency
- Low Dependency

Satisfaction Flag

Customer satisfaction was determined using review ratings.

- Rating $\geq 4 \rightarrow$ Satisfied
- Rating $< 4 \rightarrow$ Unsatisfied

Loyalty Score V1

Based on:

- Previous Purchases
- Frequency of Purchases
- Subscription Status

Loyalty Score V2

Based on:

- Review Rating
- Previous Purchases
- Promo Dependency Score

Two competing loyalty definitions were created and compared as required by the project.

7. Customer Segmentation Results

Customer Value Tier

Tier	Customers
High Value	977
Medium Value	1,905
Low Value	1,018

Loyalty Score V2

Loyalty Level	Customers
High Loyalty	1,043
Medium Loyalty	1,879
Low Loyalty	978

Satisfaction Analysis

Status	Customers
Satisfied	1,634
Unsatisfied	2,266

Promotional Dependency

Dependency Level Customers

Low Dependency 2,223

High Dependency 1,677

8. SQL Analysis

Business-focused SQL queries were used to answer key questions.

Customer Value Distribution

Most customers belong to the Medium Value segment.

Revenue by Customer Value Tier

Medium Value customers generated the highest total revenue due to their larger population size.

Loyalty Distribution

The majority of customers fall within the Medium Loyalty segment.

Satisfaction Analysis

More customers are classified as Unsatisfied than Satisfied.

Revenue by Category

Clothing generated the highest revenue.

Top Revenue Locations

Top-performing locations include:

- Montana
- Illinois
- California
- Idaho
- Nevada

Ideal Customer Segment

Ideal customers were defined as:

- High Value

- High Loyalty
- Low Promotional Dependency

A total of 356 ideal customers were identified.

9. Power BI Dashboard

A four-page interactive dashboard was developed.

Page 1: Executive Overview

Visuals Included:

- Total Revenue KPI
- Total Customers KPI
- Average Purchase Amount KPI
- Revenue by Category
- Revenue by Gender

Page 2: Customer Segmentation

Visuals Included:

- Customer Value Tier
- Loyalty Level V2
- Satisfaction Analysis
- Promotional Dependency Analysis

Page 3: Customer Behavior Analysis

Visuals Included:

- Purchase Frequency Distribution
- Payment Method Preference
- Revenue by Season
- Category Preference by Gender

Page 4: Business Recommendations

Visuals Included:

- KPI Summary
 - Strategic Business Recommendations
-

10. Key Findings

1. Clothing is the highest revenue-generating category.
 2. Medium Value customers form the largest customer segment.
 3. Approximately 43% of customers are highly dependent on promotions.
 4. More customers are unsatisfied than satisfied.
 5. Revenue remains relatively stable across seasons.
 6. Subscription status does not significantly influence spending behavior.
 7. A valuable segment of 356 ideal customers was identified.
-

11. Business Recommendations

1. Prioritize retention efforts for High Value customers.
 2. Develop loyalty programs that convert Medium Value customers into High Value customers.
 3. Reduce excessive reliance on discounts and promotions.
 4. Improve customer satisfaction through better service and personalized experiences.
 5. Increase marketing investment in high-performing categories such as Clothing and Accessories.
 6. Implement retention campaigns targeting low-frequency buyers.
 7. Focus geographic marketing efforts on top-performing states.
-

12. Conclusion

This project successfully analyzed customer purchasing behavior using Python, SQL, and Power BI. Through feature engineering, customer segmentation, SQL-based business analysis, and dashboard development, valuable insights were generated regarding customer value, loyalty, satisfaction, and promotional dependency.

The findings provide actionable recommendations that can help businesses improve customer retention, optimize marketing strategies, and increase profitability. The project demonstrates the complete end-to-end workflow of a modern data analytics solution, from raw data to business decision-making.